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Contribution Journal

Brian's Contribution

I completed the first code cell in the lab, including importing the libraries, creating the synthetic dataset, and adding comments to explain the code. This helped me prepare the dataset for the rest of the lab. It helped me to understand how to set up data for machine learning models. I also worked on the knowledge check questions. We discussed the answers, and I went through the concepts. I focused on understanding overfitting/underfitting and bias-variance trade-offs. Overall, I contributed to the coding and knowledge check sections of the assignment.

Toan's Contribution

I completed the polynomial regression section of the lab, where I worked with polynomial degrees 1, 4, and 15 to compare how each model fits the dataset. I examined the graphs and analyzed how changing the polynomial degree affected the model's performance and accuracy. By comparing the results, I was able to identify the differences between underfitting and overfitting. For example, I noticed that lower-degree models were too simple to capture the patterns in the data, while higher-degree models were more complex and could fit the data too closely. This helped me better understand how polynomial degrees affect the shape of the regression curve and why choosing the right degree is important for building a good model.

I also contributed to the knowledge check questions by discussing the answers and reviewing the main concepts we learned throughout the lab. I focused on understanding the

reasons behind each answer, especially the relationship between model complexity, underfitting, and overfitting. I also compared the different models to understand why some performed better than others and how these differences could affect the results. Through this process, I was able to connect what I learned from the graphs to the concepts discussed in the knowledge check questions. Overall, my main contributions were completing the polynomial regression section, analyzing the results, and participating in the knowledge check discussions.

Montserrat's Contribution

I completed the learning curves section of the lab. I worked with the graphs comparing training scores and cross-validation scores to understand how the models performed. I examined the differences between the scores to identify signs of overfitting and underfitting. This helped me understand how learning curves can be used to evaluate machine learning models. I also worked on the knowledge check questions by discussing the answers and reviewing the concepts from the lab. I focused on understanding what the graphs showed and how they helped identify model performance. Overall, I contributed to the learning curves section and the knowledge check questions.